FairRound field experiment: hypotheses

Pre-registration note · Oct 29, 2026 · @Amiran

Abstract. This note pre-registers a field experiment on FairRound, an on-chain launch and governance system live on Arbitrum One. From 31 October 2026 the system runs in public, unchanged and with its free entry open, and is then given an exit: a token that can only be created by burning credits released through a holder vote, traded in a small announced pool. We state four hypotheses before any data is collected: that unfair players try to capture more than their assigned share; that fair outcomes depend on diverse and active participation; that exploiters still contribute to the system; and that scale improves allocation. We also set out how individual users' arrival patterns will be compared with those of automated wallets, and which design defences we expect to hold, slow down or fail against automation. Inspired by Silvio Gesell's Freigeld and Freiland, the study asks whether procedural participation can act as a filter that people value and machines do not.

Keywords: mechanism design, token launch, on-chain governance, Sybil resistance, bots, field experiment, Gesell, Freigeld

JEL codes: D44, D47, D71, E42, G23

Purpose and context

This research tests how modern exploiters behave on a live on-chain system: how individual users arrive compared with automated proxies, and which design choices stop machines from exploiting the system and which become useless against them. The FairRound launch from 31 October 2026 is the field site, and the hypotheses below are fixed before it starts so the results can't be reinterpreted afterwards.

Research questions

1. How does the arrival pattern of individual users differ from that of automated wallets (timing, bursts, daily rhythm)?
2. Which defences in FairRound's design hold against automation, which only slow it down, and which are bypassed entirely?
3. Who captures value when machines take part, and does the system still benefit from them?
4. Does scale improve allocation: do bigger listings and more participants spread units and payouts more fairly?

The frame is Silvio Gesell's *Die natürliche Wirtschaftsordnung durch Freiland und Freigeld*:

- **Freigeld:** money should circulate, not be hoarded. UNRIG is free, can't leave the system, and is only useful when spent on bids.
- **Freiland:** the value of a shared asset belongs to the community. Each FairRound listing has a treasury its holders control by vote.

The historical reference point is the Wörgl experiment (1932–33), Gesell's money tested in one Austrian town and stopped by the central bank. Unlike Wörgl, every action here is recorded on a public chain, so behaviour can be measured, not only reported.

The setting

The experiment runs on FairRound's live contracts on Arbitrum One, in three phases, with an exit to a public pool only in the last one.

Fixed rules (the "assigned share")

- Each verified wallet can claim 1,000 UNRIG once; at most 50 new wallets are verified per hour, and each must hold 0.0083 ETH.
- One active bid per wallet per listing; at most 100 bids per listing, and a higher new bid evicts the lowest.
- Every unit filled in a round pays the same price; 99% of what is paid becomes the listing's capital, 1% goes to the host.
- Holders elect a project lead once half a listing is sold; treasury payments need a holder vote (3 days, 20% quorum), then a 5-day delay, capped at 5% of capital per 30-day term.
- UNRIG can't move between wallets. Only UNRIG paid out by a voted treasury payment or won in a treasury sale can be burned into vUNRIG, which trades freely.

Phases

1. **Baseline (before 31 October):** snapshot of wallets, holdings, bids and treasuries as they stand.
2. **Closed loop (31 October to about 17 November):** the founder runs one listing's full lifecycle with three wallets (The Fellowship), deploys vUNRIG, and the first voted payment is executed and converted.
3. **Open exit (from about 17 November):** a vUNRIG pool opens, capped at about $20 of liquidity and announced in advance; the free claim stays open throughout, so its weak points are tested in the open.

Defences under test

Each defence is listed with the way an automated exploiter could get around it, so the data can show which ones held, which only slowed machines down, and which were useless.

Defence	Meant to stop	How a machine could get around it	Sign it failed
Bot check on the website	Scripted sign-ups	Call the contracts directly and never open the site	Chain actions with no matching site event
Wallet must hold 0.0083 ETH to verify	Cheap mass wallets	Pass the same ETH from wallet to wallet; it is checked only at verification	Verified wallets funded in a chain by one source
At most 50 verifications an hour	Floods of new wallets	Patience: 1,200 a day; take slots right after each hourly reset	Hours at the cap; share of slots taken by clusters
One bid per wallet; 100 bids per book, lowest evicted	One player dominating a round	Many wallets; outbid to push site users out of full books	Site users' bids evicted by non-site wallets
Same price for everyone filled	Front-running inside a round	Bid in the last seconds with knowledge of the book	Bids clustered right before round end
UNRIG can't leave; vUNRIG only from voted payouts	Selling free claims	Win a listing's vote with many wallets, then pay themselves	A cluster electing the lead and receiving payouts
Minimum listing size for vUNRIG (9 units)	Buying a tiny listing outright	9 units is cheap; capture a small listing	vUNRIG minted from listings held mostly by one cluster
5% of capital per term, 3-day vote, 5-day delay, recall	Draining a treasury fast	Slow, steady extraction over several terms	Repeated payouts to the same side
Pool capped at about $20, announced	Large losses	Drain it within minutes of opening	Time from opening to empty; who emptied it

The minimum listing size of 9 units is a deliberate weak point: it lets the founder's own test run work, and the data will show whether machines use it.

Hypotheses

Each assumption is stated as the founder wrote it, then turned into something the chain data can confirm or refute.

H1. Unfair players try to capture more than their assigned share

Prediction: some participants will run several wallets (or bots) to take more than one person's share of free UNRIG, units, votes or pool liquidity.

Measures

- Wallet clusters: wallets funded from the same source, verified in bursts, or bidding with the same timing and amounts.
- Share of units, votes and treasury payouts held by clusters, against their share if each cluster counted as one person.
- Pressure on the limits: hours where verifications hit the cap of 50, bids evicted from full books.
- In phase 3: share of the pool taken by addresses that never used the website (chain activity with no matching analytics session).

Supported if at least one cluster holds clearly more units, votes or payouts than one wallet's share. **Refuted if** no cluster stands out after the pool opens.

H2. A fair game comes only from diversity and active participation

Prediction: listings with more distinct holders and higher turnout end up with outcomes spread more evenly; listings with few active holders get captured.

Measures

- Per listing: distinct bidders per round, concentration of unit holdings (top-holder share and Gini), election and treasury-vote turnout, how often quorum is met.
- Outcomes: whether the project lead was elected by a single cluster, and where treasury payments went (to independent recipients or to the proposer's own side).

Supported if captured outcomes appear mainly where holders are few or passive.
Refuted if capture happens just as often in diverse, active listings, or not at all.

H3. Exploiters still contribute to the community, knowingly or not

Prediction: what exploiters put in (capital, fees, settled rounds, burned UNRIG, stress on the rules) is measurable and may outweigh what they take out.

Measures

- Contribution by suspected clusters: UNRIG they paid into listing capital, host fees, rounds they settled, elections they opened, UNRIG they burned for vUNRIG.
- Extraction by the same clusters: treasury payouts received, vUNRIG sold into the pool.
- Net effect = contribution minus extraction, plus any rule weakness their behaviour revealed.

Supported if the net effect is positive or their activity exposes fixes the system adopts. **Refuted if** exploiters take out clearly more than they put in and reveal nothing new.

H4. Scale is a factor for better allocation

Prediction: the larger the listing and the more participants it has, the more evenly units and treasury payouts are spread, and the harder it is for any one cluster (human or automated) to dominate.

Measures

- Allocation quality per listing: top-holder share, Gini of unit holdings, share held by clusters, share of payouts to the proposer's side.
- Scale per listing: total units (from 9 to 1,000 and more), distinct bidders, distinct holders.
- Cost of capture: UNRIG and number of wallets a cluster would need for a majority, compared across listing sizes.
- Over time: the same measures at each reading date as participation grows.

Supported if allocation quality improves steadily with listing size and participant count. **Refuted if** large listings are captured as easily as small ones, or if quality stops improving above some size (a threshold worth reporting either way).

Data and analysis plan

All data comes from two sources, the public chain and aggregate site analytics, and is read at fixed dates rather than whenever something interesting happens.

Source	What it gives	Used for
Arbitrum One events (verify, claim, bid, settle, elections, treasury votes, burns, pool swaps)	Every action, its wallet and time	H1, H2, H3
Google Analytics events (wallet_connect, verify_wallet, claim_unrig, place_bid and others), counts only	How many actions came through the website	Humans versus scripts (H1)
The site's on-chain activity page	Funnel and daily counts	Cross-check of totals

Classifying clusters. A wallet joins a cluster only on observable links: same funding source, verification within the same few minutes, or repeated identical bids. The rules are fixed now and applied the same way to every wallet. Unclear cases stay unclustered.

Humans versus scripts. For each action type, compare chain counts with website counts per day. Actions on chain with no matching site event are counted as "not through the site", not labelled bots by name.

Arrival patterns. For every new wallet and every bid, record the time between consecutive arrivals, the hour-of-day profile and how bursty the stream is (the spread of gaps against their average). Individual users are expected to arrive irregularly, following daily rhythms in their time zones; automated wallets are expected either to arrive at near-fixed intervals or in bursts right after a limit resets. The same contact-centre arrival analysis used for staffing is applied here, per day and per hour.

Reading dates. Baseline on 31 October, before the first experiment transaction; end of phase 2 (about 17 November); then 7 and 30 days after the pool opens. Results are reported for all four dates, including the ones that don't support the hypotheses.

Limits, disclosures and ethics

The results describe this system and these participants, not economies in general, and the founder is also a participant.

- **Founder's own wallets.** The founder runs the three Fellowship wallets and the owner wallet. They are listed publicly and excluded from every cluster and capture measure.
- **Small, self-selected sample.** Participants found FairRound through events and social media; expect tens or low hundreds of wallets, too few for strong statistics. Findings are reported as descriptive patterns.

- **Test credits, not wages.** UNRIG is free and has no value; vUNRIG has a market only through a pool of about $20. Behaviour with real money may differ.
- **Uncertain labels.** A cluster may be one person with several wallets or several people sharing a funder; "not through the site" may be a script or a person using a block explorer.
- **Founder influence.** The founder announces the pool and can pause new verifications. Every such action is dated and announced, and recorded in this doc.
- **Privacy.** Only aggregate results are published. No wallet is named as an exploiter, analytics never records wallet addresses or amounts, and nobody is contacted on the basis of this analysis.
- **No promise of value.** UNRIG and vUNRIG are demo tokens. No result is presented as a price or profit signal.